

## HALLANDS VADERO, Sweden

WMO: 026050

Lat: 56.4496N

Lon: 12.5453E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.9       | 22.0 | 10.6                          | 9.5 | 23.4 | 15.0 | 11.8 | 23.9 | 40.2                  | 41.2 | 36.4 | 40.8 | 11.4                  | 60   | 0.709 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.7                    | 76.0            | 66.0 | 73.3 | 65.1 | 70.9 | 63.9 | 68.0                | 73.4 | 66.5 | 71.4 | 65.0 | 69.6 | 9.7                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.9                            | 95.6 | 70.6 | 64.5 | 90.9 | 69.4 | 62.9 | 86.0 | 68.1 | 32.3          | 73.5 | 31.2 | 71.6 | 30.0 | 69.7 | 72.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 33.7              | 30.2 | 27.4 | 17.3                       | 81.1 | 4.4                | 2.9 | 14.1                                               | 83.2 | 11.6       | 85.0 | 9.1        | 86.6 | 5.9        | 88.7 |
|                   |      |      | 15.7                       | 70.1 | 4.0                | 1.6 | 12.8                                               | 71.2 | 10.5       | 72.2 | 8.3        | 73.1 | 5.4        | 74.2 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 48.4   | 33.5  | 33.9  | 37.3  | 44.8  | 53.8  | 59.7  | 64.5  | 64.1  | 58.3  | 50.3  | 42.6  | 37.0 |
|                                                                                  | DBStd       | 12.19  | 5.85  | 5.31  | 4.85  | 5.12  | 5.18  | 3.99  | 4.04  | 3.56  | 3.93  | 4.65  | 5.37  | 6.25 |
|                                                                                  | HDD50       | 2228   | 510   | 452   | 393   | 173   | 17    | 0     | 0     | 0     | 0     | 54    | 225   | 404  |
|                                                                                  | HDD65       | 6147   | 975   | 872   | 857   | 607   | 351   | 165   | 59    | 61    | 202   | 456   | 673   | 869  |
|                                                                                  | CDD50       | 1643   | 0     | 0     | 0     | 16    | 134   | 292   | 449   | 436   | 250   | 63    | 3     | 0    |
|                                                                                  | CDD65       | 89     | 0     | 0     | 0     | 0     | 3     | 7     | 44    | 32    | 3     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 184    | 0     | 0     | 0     | 1     | 17    | 13    | 97    | 54    | 2     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 15     | 0     | 0     | 0     | 0     | 3     | 1     | 9     | 2     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 13.7   | 14.6  | 14.5  | 13.0  | 11.6  | 11.7  | 12.9  | 12.1  | 13.2  | 14.3  | 15.2  | 15.4  | 15.7 |
| Precipitation                                                                    | PrecAvg     | 28.5   | 2.2   | 1.8   | 1.5   | 1.4   | 2.0   | 2.8   | 3.2   | 3.5   | 2.6   | 2.7   | 2.3   | 2.4  |
|                                                                                  | PrecMax     | 36.5   | 4.6   | 3.9   | 3.0   | 3.1   | 4.9   | 5.1   | 8.3   | 7.8   | 5.0   | 4.7   | 4.5   | 4.4  |
|                                                                                  | PrecMin     | 20.8   | 0.2   | 0.2   | 0.4   | 0.2   | 0.7   | 0.2   | 0.1   | 1.1   | 0.6   | 0.5   | 0.5   | 0.6  |
|                                                                                  | PrecStd     | 3.9    | 1.1   | 0.7   | 0.7   | 0.8   | 1.0   | 1.3   | 1.8   | 1.5   | 1.2   | 1.1   | 1.0   | 0.9  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 45.9  | 43.9  | 52.7  | 65.7  | 75.8  | 75.6  | 80.8  | 78.5  | 72.7  | 61.6  | 54.0  | 49.1 |
|                                                                                  |             | MCWB   | 44.5  | 41.6  | 45.8  | 54.6  | 61.8  | 64.4  | 68.3  | 66.8  | 63.4  | 57.2  | 52.8  | 47.7 |
|                                                                                  | 2%          | DB     | 44.0  | 42.3  | 47.0  | 59.7  | 68.8  | 72.1  | 76.7  | 75.3  | 68.7  | 59.0  | 52.1  | 47.2 |
|                                                                                  |             | MCWB   | 42.5  | 40.6  | 43.0  | 50.9  | 58.7  | 63.2  | 67.4  | 65.4  | 61.8  | 55.9  | 50.8  | 45.9 |
|                                                                                  | 5%          | DB     | 42.2  | 41.3  | 44.5  | 55.7  | 64.8  | 69.1  | 73.7  | 72.4  | 65.7  | 57.5  | 50.9  | 45.9 |
|                                                                                  |             | MCWB   | 40.8  | 39.9  | 41.8  | 49.0  | 56.9  | 61.6  | 66.2  | 64.9  | 60.3  | 54.9  | 49.3  | 44.6 |
|                                                                                  | 10%         | DB     | 40.9  | 40.3  | 42.8  | 52.0  | 61.4  | 66.2  | 71.2  | 69.8  | 63.7  | 56.1  | 49.5  | 44.6 |
|                                                                                  |             | MCWB   | 39.7  | 39.0  | 40.9  | 47.1  | 55.4  | 60.3  | 64.7  | 63.9  | 59.1  | 53.7  | 47.6  | 43.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.1  | 42.3  | 46.4  | 56.7  | 63.6  | 67.5  | 70.5  | 69.4  | 65.4  | 58.4  | 53.1  | 47.6 |
|                                                                                  |             | MCDB   | 44.7  | 43.0  | 51.7  | 65.0  | 73.0  | 73.1  | 77.3  | 74.4  | 70.0  | 60.3  | 53.5  | 48.0 |
|                                                                                  | 2%          | WB     | 42.1  | 41.1  | 43.8  | 52.2  | 60.5  | 64.9  | 68.5  | 67.5  | 63.0  | 56.6  | 51.2  | 45.9 |
|                                                                                  |             | MCDB   | 42.8  | 41.9  | 46.3  | 57.3  | 66.1  | 70.1  | 74.5  | 72.3  | 67.0  | 58.3  | 51.8  | 46.4 |
|                                                                                  | 5%          | WB     | 40.9  | 40.2  | 42.3  | 49.8  | 58.4  | 62.9  | 67.2  | 66.0  | 61.3  | 55.4  | 49.7  | 44.5 |
|                                                                                  |             | MCDB   | 41.6  | 41.1  | 43.9  | 54.1  | 63.6  | 67.2  | 72.5  | 70.4  | 64.8  | 57.1  | 50.7  | 45.2 |
|                                                                                  | 10%         | WB     | 39.7  | 39.2  | 41.2  | 47.8  | 56.4  | 61.2  | 65.7  | 64.5  | 59.9  | 54.2  | 48.0  | 43.2 |
|                                                                                  |             | MCDB   | 40.6  | 40.2  | 42.5  | 51.2  | 60.4  | 65.5  | 70.3  | 68.9  | 63.1  | 56.0  | 49.3  | 44.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.8   | 4.8   | 6.6   | 9.3   | 10.0  | 8.9   | 8.7   | 8.4   | 7.5   | 5.8   | 4.6   | 4.7  |
|                                                                                  |             | MCDBR  | 4.2   | 4.6   | 8.7   | 15.5  | 16.4  | 13.9  | 13.6  | 13.4  | 10.8  | 6.7   | 4.7   | 4.1  |
|                                                                                  |             | MCWBR  | 4.7   | 4.0   | 6.3   | 9.7   | 10.0  | 8.9   | 8.2   | 7.9   | 6.9   | 5.4   | 5.0   | 4.6  |
|                                                                                  | 5% WB       | MCDBR  | 4.3   | 4.3   | 7.9   | 13.2  | 13.8  | 12.0  | 12.0  | 10.8  | 9.5   | 6.1   | 4.6   | 4.3  |
| MCWBR                                                                            |             | 4.7    | 4.0   | 6.0   | 8.7   | 9.2   | 8.3   | 7.8   | 7.6   | 6.9   | 5.7   | 5.0   | 4.8   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.310  | 0.334 | 0.354 | 0.389 | 0.385 | 0.377 | 0.400 | 0.401 | 0.370 | 0.354 | 0.333 | 0.300 |      |
|                                                                                  | taud        | 2.275  | 2.307 | 2.329 | 2.294 | 2.376 | 2.425 | 2.376 | 2.380 | 2.445 | 2.403 | 2.321 | 2.249 |      |
|                                                                                  | Ebn at Noon | 183    | 228   | 255   | 262   | 271   | 275   | 265   | 256   | 248   | 219   | 173   | 153   |      |
|                                                                                  | Edh at Noon | 17     | 24    | 30    | 36    | 35    | 34    | 35    | 33    | 27    | 22    | 16    | 13    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 131    | 316   | 746   | 1256  | 1640  | 1722  | 1597  | 1285  | 903   | 434   | 163   | 89    |      |
|                                                                                  | RadStd      | 25     | 51    | 76    | 169   | 157   | 149   | 192   | 110   | 104   | 65    | 30    | 10    |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (64 neighbors) |  | +0.76 | N/A     | N/A    | N/A     | N/A   | N/A   | -216        | -310  | N/A   | N/A   |

Nomenclature: See separate page